



Chapter 4 Multivariate Normal Population Statistical Inference



Content

- § 4.1 Univariate Situation Review
- § 4.2 Single Population Mean Inference
- § 4.3 Single Population Mean Component Structural Relationship Test
- § 4.4 Two Population Mean Comparative Inference
- § 4.5 Two Population Mean Component Structural Relationship Test
- § 4.6 Multiple Population Mean Comparative Test
(Multivariate Analysis of Variance)
- § 4.7 Population Correlation Coefficient Inference



§4.1 Univariate Situation Review

- §4.1.1 Single Normal Population Mean Confidence Interval
- §4.1.2 Difference between Two Normal Populations Confidence Interval
- §4.1.3 Relationship between hypothesis test and confidence interval
- §4.1.4 Multiple Population Mean Comparative Test

§ 4.1 Univariate Situation Review

➤ —. Confidence Interval

Let $\hat{\theta}_L = \hat{\theta}_L(x_1, \dots, x_n)$ and $\hat{\theta}_U = \hat{\theta}_U(x_1, \dots, x_n)$ be the lower and upper bounds of θ 's interval estimation. respectively

θ is estimated between $\hat{\theta}_L$ and $\hat{\theta}_U$, $\hat{\theta}_L \leq \theta \leq \hat{\theta}_U$.

Confidence degree : the probability that θ is included in stochastic interval $(\hat{\theta}_L, \hat{\theta}_U)$, denoted by $1 - \alpha$.

➤ Given **confidence level** α , we have

$$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha$$

where $(\hat{\theta}_L, \hat{\theta}_U)$ is the confidence interval of unknown parameter θ with confidence degree $1 - \alpha$.

§4.1.1 Single Normal Population Mean Confidence Interval

- Suppose $(x_1, \dots, x_n)$ is randomly chosen from normal population $N(\mu, \sigma^2)$. Given confidence level α , find the confidence interval of μ with confidence degree $1 - \alpha$.
- σ^2 is known, the low and upper confidence bounds of μ :

$$\bar{x} \pm u_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $u_{\alpha/2}$ is the $\alpha/2$ quantile of $N(0,1)$.

- σ^2 is unknown, the low and upper confidence bounds of μ :

$$\bar{x} \pm t_{\alpha/2} (n-1) \frac{s}{\sqrt{n}}$$

§4.1.2 Confidence Interval of the difference between Two Normal Populations

➤ Suppose $\bar{x} = \frac{1}{n} \sum_{i=1}^{n_1} x_i$ and $\bar{y} = \frac{1}{n_2} \sum_{i=1}^{n_2} y_i$ are respectively the sample means of two independent normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$.

■ σ_1^2 and σ_2^2 are known, the $1 - \alpha$ confidence interval of $\mu_1 - \mu_2$:

$$\bar{x} - \bar{y} \pm u_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

■ $\sigma_1^2 = \sigma_2^2 = \sigma^2$ is unknown, $1 - \alpha$ confidence interval of $\mu_1 - \mu_2$:

$$\bar{x} - \bar{y} \pm t_{\alpha/2}(n_1 + n_2 - 2) s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Where $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$ is an unbiased estimator of σ^2

二. Hypothesis Test

- Null Hypothesis and Alternative Hypothesis:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

- Rejection rule :

If $|\bar{x} - \mu_0| \geq c$, then H_0 is rejected, where c is critical value.

- Type I Error: To reject H_0 when null hypothesis is true;
- Type II Error: To accept H_0 when null hypothesis is not true.
- Neiman Pearson Principle: the probability α of making Type I Error should be controlled .

Hypothesis test for single normal population

- Consider the Hypothesis test:

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$$

- Assuming $(x_1, x_2, \dots, x_n)$ is a sample from $N(\mu, \sigma^2)$, where σ^2 is known, given the significance level α , the test statistic

$$u = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- Under H_0 . Then the reject rule is :

$$\text{To reject } H_0 \text{ if } u = \left| \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \right| \geq u_{\alpha/2}$$

§4.1.3 Relationship between hypothesis test and confidence interval

➤ Accept $H_0 : \mu = \mu_0$ at significance level α , equivalent to

$$\left| \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \right| \leq u_{\alpha/2}$$

i.s.
$$\bar{x} - u_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + u_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

μ_0 is included in the $1 - \alpha$ confidence interval of μ .

P- value

- **U** denotes test statistics;
u denotes the value of test statistics ;
 $u_{\alpha/2}$ denotes $\alpha / 2$ quantile of **U**;
- Rejection rule : If $|u| \geq u_{\alpha/2}$, reject H_0 ;
- **P-value rule** :

$$p = P(|U| \geq |u|) \leq P(|U| \geq u_{\alpha/2}) = \alpha$$

- If $p \leq \alpha$, then reject H_0 .
- P-value is automatically output on some software, such as SAS,SPSS.

§4.1.4 Multiple Population Mean Comparative Test

- Suppose $x_{i1}, \dots, x_{in}$, $i = 1, \dots, k$ is a sample randomly chosen from population $\pi_i \sim N(\mu_i, \sigma^2)$.

Now we want to test

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k, \quad H_1 : \mu_i \neq \mu_j$$

where at least one couple elements satisfy $i \neq j$

➡ Suppose $x_{i1}, \dots, x_{in_i}$, $i = 1, \dots, k$ is a sample of population

$$\pi_i \sim N(\mu_i, \sigma^2)$$

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2, \quad SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

$$SSTR = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2 \quad \text{where, } \bar{x} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} x_{ij}$$

$$n = \sum_{i=1}^k n_i \quad \bar{x}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} x_{ij} \quad (i = 1, 2, \dots, k). \quad \text{Then it is easy to verify that}$$

total deviation sum of square: $SST = SSE + SSTR$,

sum of squares of errors

sum of squares of
total regression

判定系数 $R^2 = \frac{SSTR}{SST}$, the larger for R^2 , more significant for the difference among the groups

- It is easy to verify that when H_0 is true, $\frac{SSE}{\sigma^2} \sim \chi^2(n - k)$,
 $\frac{SSTR}{\sigma^2} \sim \chi^2(k - 1)$
- Construct test statistic

$$F = \frac{SSTR / (k - 1)}{SSE / (n - k)} \quad (4.1.24)$$

When H_0 is true, $F \sim F(k - 1, n - k)$, the rejection rule is that if $F \geq F_\alpha(k - 1, n - k)$, then reject H_0 .

Theorem Under the above hypotheses:

(1) $\frac{SSE}{\sigma^2} \sim \chi^2(n-k);$

(2) When H_0 is true, $\frac{SSTR}{\sigma^2} \sim \chi^2(k-1), \frac{SST}{\sigma^2} \sim \chi^2(n-1);$

(3) SSE and $SSTR$ are mutually independent.

When H_0 is true, test statistic:

$$F = \frac{SSTR / (k-1)}{SSE / (n-k)} \sim F(k-1, n-k)$$

**total regression
sum of squares**

**error sum of
squares**

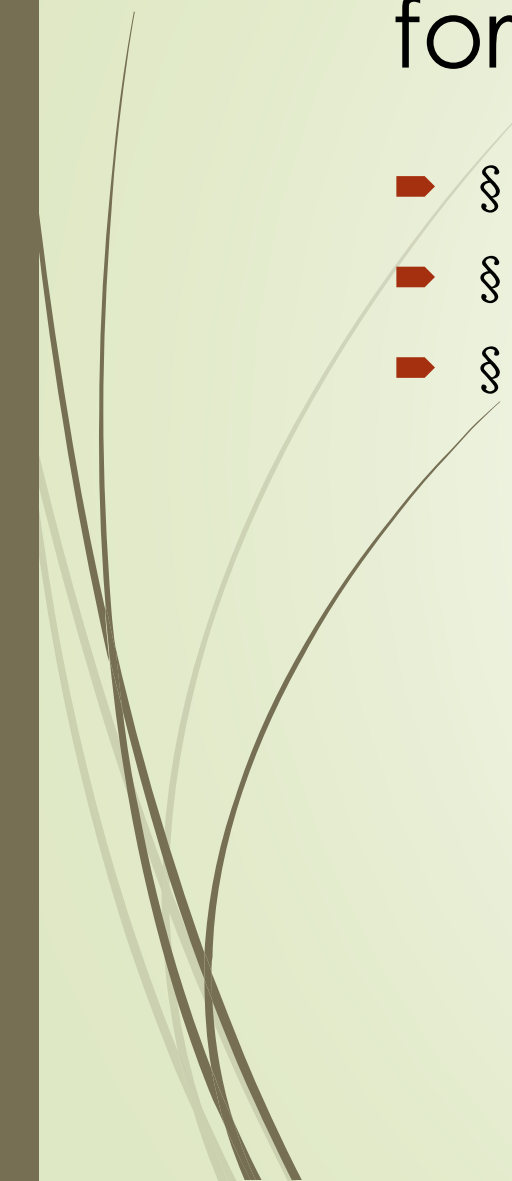


Assignment: Prove(4.1.23) in P75.





§ 4.2 Single Population Mean Inference for multivariate situation

- § 4.2.1 Mean Vector Test
 - § 4.2.2 Confidence Region
 - § 4.2.3 Simultaneous Confidence Interval
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§ 4.2.1 Mean Vector Test

- Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is a sample of population $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} > 0, n > p$. The hypotheses of the test:

$$H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0, H_1: \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

- 1. $\boldsymbol{\Sigma}$ is known: Test statistic:

$$T_0^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$$

rejection rule: If $T_0^2 \geq \chi_{\alpha}^2(p)$, then reject H_0 .

- 2. $\boldsymbol{\Sigma}$ is unknown: Test statistic

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \quad (4.2.4)$$

is called **Hotelling T^2 Statistic** (霍特林 T^2 统计量).

- **T^2 Statistic** is a generalization of t statistics for multivariate situation.

t statistics: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t(n - 1)$.

► Hotelling Distribution :

$\mathbf{x} \sim N_p(\mathbf{0}, \Sigma)$, $\mathbf{W} \sim W_p(\mathbf{n}, \Sigma)$, $\mathbf{x}$ and $\mathbf{W}$ are mutually independent, then the distribution of $T^2 = n\mathbf{x}'\mathbf{W}^{-1}\mathbf{x}$ is **Hotelling Distribution** with free degree n .

➤ Hotelling distribution T^2 is unrelated to Σ

$$T^2 = n(\Sigma^{-1/2}\mathbf{x})'(\Sigma^{-1/2}\mathbf{W}\Sigma^{-1/2})^{-1}(\Sigma^{-1/2}\mathbf{x})$$

➤ $\Sigma^{-1/2}\mathbf{x} \sim N_p(0, I)$, $\Sigma^{-1/2}\mathbf{W}\Sigma^{-1/2} \sim W_p(n, I)$.

► Properties :

$$\frac{n - p + 1}{pn} T^2(p, n) \sim F(p, n - p + 1)$$

where n is the free degree of the above Wishart Distribution.

► Hotelling Distribution :

$\mathbf{x} \sim N_p(\mathbf{0}, \Sigma)$, $\mathbf{W} \sim W_p(\mathbf{n}, \Sigma)$, $\mathbf{x}$ and $\mathbf{W}$ are mutually independent, then the distribution of $T^2 = n\mathbf{x}'\mathbf{W}^{-1}\mathbf{x}$ is **Hotelling Distribution** with free degree n .

- When $p=1$, $\mathbf{x} \sim N(0, 1)$, $W \sim \chi^2(n)$, $\mathbf{x}$ is independent of W , we obtain

$$T = \frac{x}{\sqrt{W/n}} \sim t(n), \quad T^2 = \frac{nx^2}{W} \sim T^2(1, n). \text{ So, we have}$$

$$T^2(1, n) = t^2(n).$$

- Hotelling T^2 -distribution is actually a generalization of t – distribution in multivariate case.

2. Σ is unknown:

Test statistic

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$$

$$T^2 = (n-1)[\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)]' [(n-1)\mathbf{S}]^{-1} [\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)]$$

is called **Hotelling T^2 Statistic** (霍特林 T^2 统计量) .

The above $T^2 \sim T^2(p, n-1)$, i.e., the Hoteling Distribution with free degree $n-1$.

- When H_0 is true, $\frac{n-p}{p(n-1)} T^2$ is distributed as **$F(p, n-p)$** .

$$T^2 = (n-1)[\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)]' [(n-1)\mathbf{S}]^{-1} [\sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)] \sim T^2(p, n-1)$$

- Given significance level α , the rejection rule is :

$$\text{If } T^2 \geq T_{\alpha}^2 \text{ then reject } H_0.$$

where

$$T_{\alpha}^2 = \frac{p(n-1)}{n-p} F_{\alpha}(p, n-p)$$

- **Example 4.2.1** We measure the height and chest circumference and upper arm girth of 6 two-year-old boys from a certain region, and we attain the sample data as Table 4.2.1 displays. According to former material, the mean of these 3 indexes of two-year-old boys in the city of this region is $\mu_0=(90,58,16)'$. Now we plan to test whether rural boys have the same mean vector in terms of the three indexes as urban boys. This is equivalent to the following hypothesis test problem :

$$H_0: \mu=\mu_0, \quad H_1: \mu\neq\mu_0$$

Table 4. 2. 1

Physical Measurement Data

Number	Height(cm)	C.C(cm)	U.A.G(cm)
1	78	60.6	16.5
2	76	58.1	12.5
3	92	63.2	14.5
4	81	59.0	14.0
5	81	60.8	15.5
6	84	59.5	14.0

➤ Solution: By calculation, we obtain

$$\bar{\mathbf{x}} = \begin{pmatrix} 82.0 \\ 60.2 \\ 14.5 \end{pmatrix}, \quad \bar{\mathbf{x}} - \boldsymbol{\mu}_0 = \begin{pmatrix} -8.0 \\ 2.2 \\ -1.5 \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} 31.600 & 8.040 & 0.500 \\ 8.040 & 3.172 & 1.310 \\ 0.500 & 1.310 & 1.900 \end{pmatrix}$$

$$\mathbf{S}^{-1} = (23.13848)^{-1} \begin{pmatrix} 4.3107 & -14.6210 & 8.9464 \\ -14.6210 & 59.7900 & -37.3760 \\ 8.9464 & -37.3760 & 35.5936 \end{pmatrix}$$

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0) = 6 \times 70.0741 = 420.445$$

By noticing $F_{0.01}(3,3)=29.5$ and $\frac{n-p}{p(n-1)} T^2 \sim F(p, n-p)$, we have

$$T_{0.01}^2 = \frac{3 \times 5}{3} F_{0.01}(3,3) = 147.5 < 420.445 = T^2.$$

Therefore, given the significance level $\alpha=0.01$, we should reject H_0 . This implies that the above indexes between rural and urban two-year-old boys are significantly different ($p=0.002$).

§ 4.2.2 Confidence Region of Hotelling test

For the hypotheses of the test:

$$H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$$

Steps for constructing Hypothesis testing with Hotelling T^2 Statistic:

Step 1: Construct the statistic $T^2 = n(\bar{x} - \mu_0)' S^{-1} (\bar{x} - \mu_0)$

Step 2: Determine the null distribution

$$\frac{n-p}{p(n-1)} T^2 \sim F(p, n-p)$$

Step 3: Determine the reject rule:

$$\text{Reject } H_0 \text{ if } T^2 > T_\alpha^2 \equiv \frac{p(n-1)}{n-p} F_\alpha(p, n-p).$$

Acceptance region of this test is given by

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq T_\alpha^2$$


$$P\left(n(\bar{\mathbf{x}} - \boldsymbol{\mu})' \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \leq T_{\alpha}^2\right) = 1 - \alpha$$

- Confidence region of $\boldsymbol{\mu}$ with significance level $1-\alpha$ is given by

$$\left\{ \boldsymbol{\mu} : n(\bar{\mathbf{x}} - \boldsymbol{\mu})' \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \leq T_{\alpha}^2 \right\} \quad (4.2.14)$$

When $p=1$, it is a interval; When $p=2$, it becomes an ellipse and we can draw it on a 2-D coordinate; When $p=3$, it is an ellipsoid; When $p > 3$, it turns to a hyper-ellipsoid.

- Relationship between confidence region and hypothesis test: Generally speaking, $\boldsymbol{\mu}_0$ is included in the above confidence region, if and only if $H_0: \boldsymbol{\mu}=\boldsymbol{\mu}_0$ is accepted at significance level $1-\alpha$. Therefore, hypothesis test can be taken by means of constructing confidence region.

§ 4.2.2 Simultaneous Confidence Interval

- If the conclusion is to reject H_0 when constructing the mean vector test

$$H_0: \mu = \mu_0 \quad vs \quad H_1: \mu \neq \mu_0,$$

we hope to find which components cause μ and μ_0 to be different.

Therefore, we need to test

$$H_{0i}: \mu_i = \mu_{0i} \quad vs \quad H_{1i}: \mu_i \neq \mu_{0i}.$$

- Let $\mu_0 = \mathbf{0}$ for simplicity. Then, the above tests can be summarized as

$$H_{0i}: a_i' \mu = 0 \quad vs \quad H_{1i}: a_i' \mu \neq 0$$

for $i = 1, 2, \dots, k$, where a_i represent the i -th base vector in p -dimension space.

- Judging whether the null hypotheses in the above tests are true at the same time is actually equivalent to finding the simultaneous confidence interval for $a_i' \mu$.

§ 4.2.2 Simultaneous Confidence Interval(联合置信区间)

- More generally, we are interested in the following test

$$H_0: a'\mu = 0 \quad vs \quad H_1: a'\mu \neq 0 \quad \text{for any } a \neq 0.$$

- Suppose $x_1, x_2, \dots, x_n$ are samples from population $N_p(\mu, \Sigma)$. Let $y_i = a'x_i$ for any $a \neq 0$. Then, $y_1, y_2, \dots, y_n$ are the samples from $N(a'\mu, a'\Sigma a)$ with the sample mean

$$\bar{y}_a = \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n a'x_i = a'\bar{x}$$

and the sample covariance

$$s_a^2 = \frac{1}{n-1} \sum_{i=1}^n (a'x_i - a'\bar{x})^2 = \frac{1}{n-1} a' \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})' a = a'Sa.$$

From the t statistic $t = \frac{a'\bar{x} - a'\mu}{s_a/\sqrt{n}} \sim t(n-1)$, we have the following confidence interval of $a'\mu$ with confidence level $1 - \alpha$ given by

$$(a'\bar{x} - t_{\alpha/2}(n-1)\sqrt{a'sa}/\sqrt{n}, a'\bar{x} + t_{\alpha/2}(n-1)\sqrt{a'sa}/\sqrt{n}) \quad (*)$$

§ 4.2.3 Simultaneous Confidence Interval

➡ On the other hand, we have

$$(x'y)^2 \leq (x'Bx)(y'B^{-1}y) \quad (1.8.2)$$

$$[a'(\bar{x} - \mu)]^2 \leq a'Sa(\bar{x} - \mu)'S^{-1}(\bar{x} - \mu)$$

by Cauchy Inequality, where the equality holds when $a \propto S^{-1}(\bar{x} - \mu)$.
Therefore,

$$\max_{a \neq 0} \frac{n[a'(\bar{x} - \mu)]^2}{a'Sa} = n(\bar{x} - \mu)'S^{-1}(\bar{x} - \mu) \equiv T^2(p, n-1).$$

By using $P(n(\bar{x} - \mu)'S^{-1}(\bar{x} - \mu) \leq T_\alpha^2) = 1 - \alpha$ with $T_\alpha \equiv T_\alpha(p, n-1)$,

we obtain Hotelling statistic

$$P\left\{\max_{a \neq 0} \frac{n[a'(\bar{x} - \mu)]^2}{a'Sa} \leq T_\alpha^2(p, n-1)\right\} = 1 - \alpha$$

$$P\left\{\bigcap_{a \neq 0} \frac{n|a'\bar{x} - a'\mu|^2}{a'Sa} \leq T_\alpha^2(p, n-1)\right\} = 1 - \alpha$$

§ 4.2.3 Simultaneous Confidence Interval

➡ Further, we have

$$P\left\{\bigcap_{a \neq 0} \frac{n |a'\bar{x} - a'\mu|^2}{a'Sa} \leq T_\alpha^2(p, n-1)\right\} = 1 - \alpha$$

$$P\left(\bigcap_a \left\{a'\bar{x} - T_\alpha \sqrt{a'Sa}/\sqrt{n} \leq a'\mu \leq a'\bar{x} + T_\alpha \sqrt{a'Sa}/\sqrt{n}\right\}\right) = 1 - \alpha \quad (4.2.15)$$

$$a'\bar{x} - T_\alpha \sqrt{a'Sa}/\sqrt{n} \leq a'\mu \leq a'\bar{x} + T_\alpha \sqrt{a'Sa}/\sqrt{n}. \quad (4.2.16)$$

where $T_\alpha \equiv T_\alpha(p, n-1)$. The above inequality is established for all $a \in R^p$ with probability $1-\alpha$. Then it is called the **simultaneous confidence interval**(联合置信区间) of all the linear combination $\{a'\mu, a \in R^p\}$ with confidence coefficient $1-\alpha$.

➡ For k linear combination $\{a_i'\mu, i=1,2,\dots,k\}$,

$$P\left(\bigcap_{i=1}^k \left\{a_i'\bar{x} - T_\alpha \sqrt{a_i'Sa_i}/\sqrt{n} \leq a_i'\mu \leq a_i'\bar{x} + T_\alpha \sqrt{a_i'Sa_i}/\sqrt{n}\right\}\right) \geq 1 - \alpha \quad (4.2.17)$$

Based on T^2 statistic

- When k is little, simultaneous T^2 confidence interval:

$$\mathbf{a}'_i \bar{\mathbf{x}} - T_\alpha \sqrt{\mathbf{a}'_i \mathbf{S} \mathbf{a}_i} / \sqrt{n} \leq \mathbf{a}'_i \boldsymbol{\mu} \leq \mathbf{a}'_i \bar{\mathbf{x}} + T_\alpha \sqrt{\mathbf{a}'_i \mathbf{S} \mathbf{a}_i} / \sqrt{n}, \quad i = 1, 2, \dots, k \quad (4.2.18)$$

Its confidence coefficient is usually obviously bigger than $1-\alpha$, therefore the above interval is too wide, namely low accuracy. Alternatively, we consider to use **Bonferroni (邦弗伦尼) simultaneous confidence interval (CI)**:

$$\mathbf{a}'_i \bar{\mathbf{x}} - t_{\alpha/2k}(n-1) \sqrt{\mathbf{a}'_i \mathbf{S} \mathbf{a}_i} / \sqrt{n} \leq \mathbf{a}'_i \boldsymbol{\mu} \leq \mathbf{a}'_i \bar{\mathbf{x}} + t_{\alpha/2k}(n-1) \sqrt{\mathbf{a}'_i \mathbf{S} \mathbf{a}_i} / \sqrt{n} \quad (4.2.19)$$

Its confidence coefficient is at least $1-\alpha$. Based on t statistic

- If $t_{\alpha/2k}(n-1) \leq T_\alpha$, then Bonferroni simultaneous confidence interval is narrower than T^2 interval. On this condition the former is preferable. Otherwise, if $t_{\alpha/2k}(n-1) > T_\alpha$, then Bonferroni simultaneous confidence interval is wider than T^2 interval. The latter is a better choice.

$$(\mathbf{a}'_i \bar{\mathbf{x}} - t_{\alpha/2}(n-1) \sqrt{\mathbf{a}'_i \mathbf{s} \mathbf{a}_i} / \sqrt{n}, \mathbf{a}'_i \bar{\mathbf{x}} + t_{\alpha/2}(n-1) \sqrt{\mathbf{a}'_i \mathbf{s} \mathbf{a}_i} / \sqrt{n})$$

$(1-\alpha)$ CI of $\mathbf{a}'_i \boldsymbol{\mu}$ for t test

The Bonferroni intervals for linear combination $a'_i\mu$ and the analogous T^2 intervals have the same general form

$$a'_i\bar{x} \mp (\text{critical value})\sqrt{\frac{a'_iSa}{n}}.$$

from equation (4.2.18) and (4.2.19).

$$T_\alpha^2 = \frac{p(n-1)}{n-p} F_\alpha(p, n-p)$$

Consequently, in every instance,

$$\text{Ratio of interval length} = \frac{\text{Length of Bonferroni interval}}{\text{Length of } T^2\text{-interval}} = \frac{t_{n-1}(\alpha/2k)}{\sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)}},$$

which does not depend on the random quantities $\bar{x}$ and $\mathbf{S}$. For a small number k of the specified parametric functions $a'_i\mu$ ($i = 1, \dots, k$), the Bonferroni intervals will always be shorter.

How much shorter is indicated in the following table for selected n and p .

Ratio of interval length for $1 - \alpha = 0.95$ between the Bonferroni intervals and the analogous T^2 intervals			
n	$k = p$		
	2	4	10
15	0.88	0.69	0.29
50	0.91	0.78	0.58
100	0.91	0.80	0.62
∞	0.91	0.81	0.66

- When $k \leq p$, Bonferroni simultaneous confidence interval is narrower than T^2 interval (see Example 4.2.2). Then Bonferroni method should be applied to calculate the simultaneous confidence interval of all p components of $\boldsymbol{\mu}$, $\mu_1, \mu_2, \dots, \mu_p$

- **Example 4.2.2** In order to evaluate the teaching efficiency of a certain professional training center, we randomly choose 8 trainees as a sample and they are tested for item A and item B. The data is as follows Table 4.2.2. Assume $\mathbf{x}=(x_1, x_2)'$ is distributed as bivariate normal distribution.

Table 4.2.2

Test Performance of Test A and B

Number	1	2	3	4	5	6	7	8
Performance of A: x_1	62	80	66	84	75	80	54	79
Performance of B: x_2	70	77	75	87	87	91	61	84

- **Solution:** It is known that $n=8, p=2, 1-\alpha=0.90, F_{0.10}(2,6)=3.46$,

then, $T_{0.10}^2 = \frac{2 \times 7}{6} F_{0.10}(2,6) = 8.073$. From calculation, we have

$$T_{\alpha}^2 = \frac{p(n-1)}{n-p} F_{\alpha}(p, n-p)$$

$$\bar{\mathbf{x}} = \begin{pmatrix} 72.5 \\ 79 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 112.5714 & 96.1429 \\ 96.1429 & 103.1429 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} 0.0436 & -0.0406 \\ -0.0406 & 0.0475 \end{pmatrix}$$

➤ 0.90 confidence region of μ : $\left\{ \mu : n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq T_{\alpha}^2 \right\}$

$$8 \times (72.5 - \mu_1, 79 - \mu_2) \begin{pmatrix} 0.0436 & -0.0406 \\ -0.0406 & 0.0475 \end{pmatrix} \begin{pmatrix} 72.5 - \mu_1 \\ 79 - \mu_2 \end{pmatrix} \leq 8.073$$

$$0.0436 \times (\mu_1 - 72.5)^2 - 0.0812 \times (\mu_1 - 72.5)(\mu_2 - 79) \\ + 0.0475 \times (\mu_2 - 79)^2 \leq 1.009$$

This is an ellipse region.

0.90 T^2 simultaneous CI for μ_1 and μ_2 :

$$72.5 - 2.841 \times \sqrt{112.5714 / 8} \leq \mu_1 \leq 72.5 + 2.841 \times \sqrt{112.5714 / 8}$$

$$79 - 2.841 \times \sqrt{103.1429 / 8} \leq \mu_2 \leq 79 + 2.841 \times \sqrt{103.1429 / 8}$$

i.e. $a_i' \bar{x} - T_{\alpha} \sqrt{a_i' S a_i} / \sqrt{n} \leq a_i' \mu \leq a_i' \bar{x} + T_{\alpha} \sqrt{a_i' S a_i} / \sqrt{n}, \quad i = 1, 2$

$$61.84 \leq \mu_1 \leq 83.16, \quad 68.80 \leq \mu_2 \leq 89.20$$

These two intervals are just projections of ellipse on μ_1 axis and μ_2 axis.

$$\mathbf{a}_i' \bar{\mathbf{x}} - t_{\alpha/2k} (n-1) \sqrt{\mathbf{a}_i' \mathbf{S} \mathbf{a}_i} / \sqrt{n} \leq \mathbf{a}_i' \boldsymbol{\mu} \leq \mathbf{a}_i' \bar{\mathbf{x}} + t_{\alpha/2k} (n-1) \sqrt{\mathbf{a}_i' \mathbf{S} \mathbf{a}_i} / \sqrt{n}$$

- Applying 0.90 **Bonferroni simultaneous CI** between μ_1 and μ_2 is ($t_{0.025}(7)=2.3646$, take $\mathbf{a}_1 = (1,0)'$, $\mathbf{a}_2 = (0,1)'$), we obtain

$$72.5 - 2.3646 \times \sqrt{112.5714 / 8} \leq \mu_1 \leq 72.5 + 2.3646 \times \sqrt{112.5714 / 8}$$

$$79 - 2.3646 \times \sqrt{103.1429 / 8} \leq \mu_2 \leq 79 + 2.3646 \times \sqrt{103.1429 / 8}$$

Furthermore, we have

$$63.63 \leq \mu_1 \leq 81.37, \quad 70.51 \leq \mu_2 \leq 87.49$$

- Accuracy of Bonferroni simultaneous confidence interval is higher than that of T^2 simultaneous confidence interval. Based on Bonferroni interval we can obtain rectangle confidence region with at least 0.90 confidence coefficient (see Figure 4.2.1 full line rectangle), but rectangle area is bigger than ellipse area.

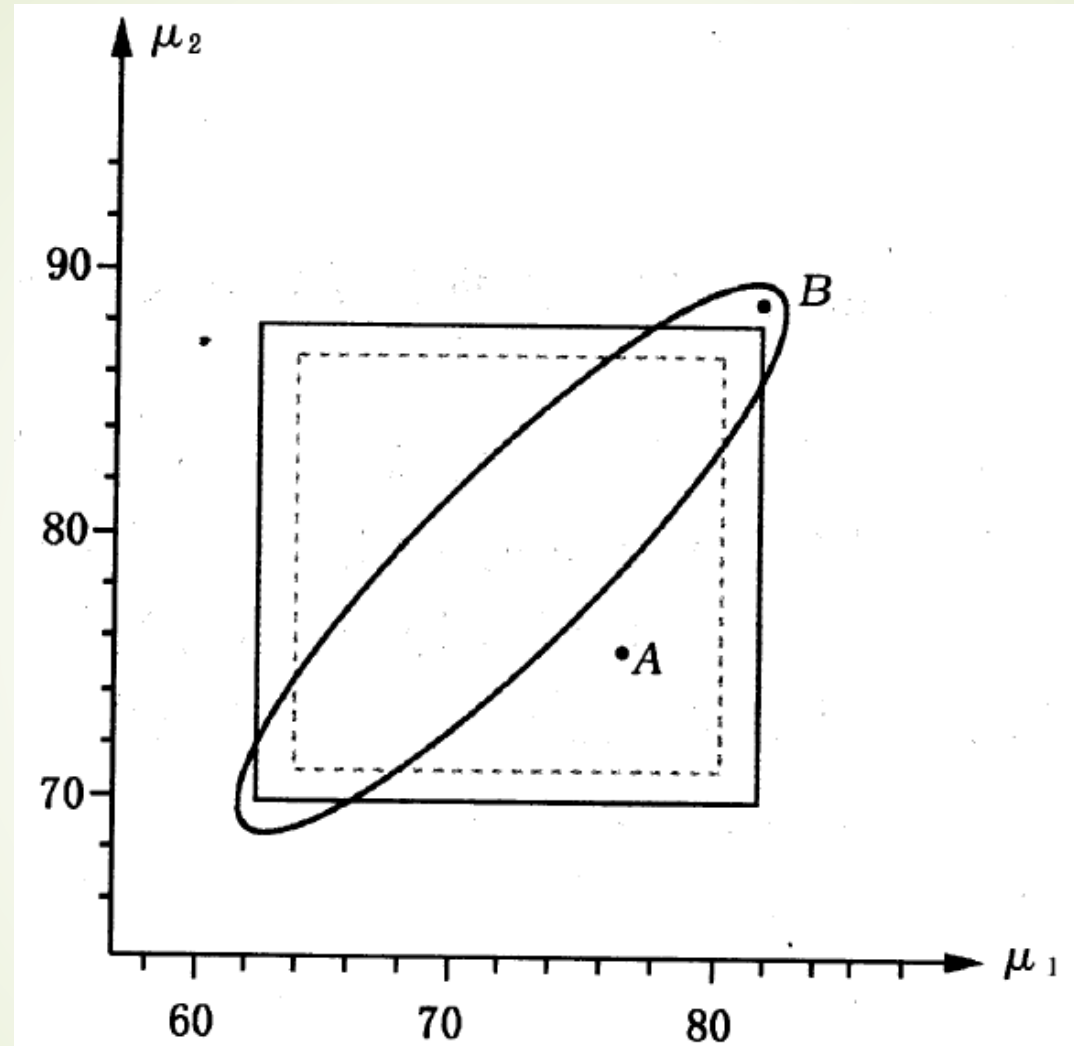


Figure 4.2.1 Confidence Ellipse and Simultaneous Confidence Interval

Hypothesis test using confidence region

- In Example 4.2.2, when $\alpha=0.10$, then test the following hypothesis

$$H_0: \boldsymbol{\mu}=\boldsymbol{\mu}_0, \quad H_1: \boldsymbol{\mu}\neq\boldsymbol{\mu}_0$$

where $\boldsymbol{\mu}=(\mu_1,\mu_2)'$, $\boldsymbol{\mu}_0=(\mu_{01},\mu_{02})'$. Then we can easily obtain the test result by using Figure 4.2.1. If test value $\boldsymbol{\mu}_0$ lies out of the ellipse in Figure 4.2.1, then reject; otherwise, then accept.

- In Figure 4.2.1, the intervals of imaginary line rectangle on μ_1 and μ_2 axis are respectively 0.90 confidence intervals of μ_1 and μ_2 . When $\boldsymbol{\mu}_0$ lies outside ellipse but inside imaginary line rectangle (like Point *A*), although test result is to reject H_0 , if we test the following hypotheses when $\alpha=0.10$

$$H_{01}: \mu_1=\mu_{01}, \quad H_{11}: \mu_1\neq\mu_{01} \quad \text{and} \quad H_{02}: \mu_2=\mu_{02}, \quad H_{12}: \mu_2\neq\mu_{02}$$

the results will both be accepting the null hypotheses; When $\boldsymbol{\mu}_0$ lies inside ellipse but outside imaginary line rectangle (like Point *B*), although test result is to accept H_0 , $H_{01}: \mu_1=\mu_{01}$ and $H_{02}: \mu_2=\mu_{02}$ will both be rejected.



Assignments: Please find the assignment file in BB.





§ 4.3 Two Population Mean Comparative Inference

- § 4.3.1 two independent samples situation
- § 4.3.2 pairing test T^2 statistic

§ 4.3.1 two independent samples situation

- Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_1}$ and $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n_2}$ are samples independently and randomly chosen from populations $N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ respectively, $\boldsymbol{\Sigma} > 0$. Hypothesis (两个独立样本的情形):

$$H_0: \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2, \quad H_1: \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$$

unbiased estimates of $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$:

$$\bar{\mathbf{x}} = \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbf{x}_i, \quad \bar{\mathbf{y}} = \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbf{y}_i$$

joint unbiased estimate of $\boldsymbol{\Sigma}$:

$$\mathbf{S}_p = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}$$

where $\mathbf{S}_1 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})'$, $\mathbf{S}_2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})'$

are covariance matrices of the samples.

Hotelling T^2 test statistic is defined as

$$T^2 = \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-1} (\bar{\mathbf{x}} - \bar{\mathbf{y}})' \mathbf{S}_p^{-1} (\bar{\mathbf{x}} - \bar{\mathbf{y}}) = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}} - \bar{\mathbf{y}})' \mathbf{S}_p^{-1} (\bar{\mathbf{x}} - \bar{\mathbf{y}}) \quad (4.3.2) \text{ Why ?}$$

When H_0 is true, $T^2 \sim T^2(p, n_1 + n_2 - 2)$. Why ? This leads to

$$\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} T^2 \sim F(p, n_1 + n_2 - p - 1) \quad (4.3.3)$$

For a given α , rejection rule is:

$$\text{If } T^2 \geq T_\alpha^2, \text{ reject } H_0. \quad (4.3.4)$$

where

$$T_\alpha^2 = \frac{p(n_1 + n_2 - 2)}{n_1 + n_2 - p - 1} F_\alpha(p, n_1 + n_2 - p - 1) \quad (4.3.5)$$

- In real application, once $H_0: \mu_1 = \mu_2$ is rejected, then we can further consider to test $H_{0i}: \mu_{1i} = \mu_{2i}$ under the same significance level including all the $i(1 \leq i \leq p)$ to judge the existence of components that play significant role in rejecting H_0 :

$$\mu_1 = \mu_2. \quad \mathbf{a}'\bar{\mathbf{x}} - T_\alpha \sqrt{\mathbf{a}'\mathbf{S}\mathbf{a}} / \sqrt{n} \leq \mathbf{a}'\boldsymbol{\mu} \leq \mathbf{a}'\bar{\mathbf{x}} + T_\alpha \sqrt{\mathbf{a}'\mathbf{S}\mathbf{a}} / \sqrt{n}. \quad (4.2.16)$$

- Similar with (4.2.16), $1-\alpha$ simultaneous CI of $\{\mathbf{a}'(\mu_1 - \mu_2), \mathbf{a} \in R^p\}$ is

$$\mathbf{a}'(\bar{\mathbf{x}} - \bar{\mathbf{y}}) - T_\alpha \sqrt{\frac{n_1 + n_2}{n_1 n_2}} \sqrt{\mathbf{a}'\mathbf{S}_p\mathbf{a}} \leq \mathbf{a}'(\mu_1 - \mu_2) \leq \mathbf{a}'(\bar{\mathbf{x}} - \bar{\mathbf{y}}) + T_\alpha \sqrt{\frac{n_1 + n_2}{n_1 n_2}} \sqrt{\mathbf{a}'\mathbf{S}_p\mathbf{a}} \quad (4.3.6)$$

- T_α is defined in (4.3.5). $\left(\frac{1}{n_1} + \frac{1}{n_2}\right)^{-1/2} [\mathbf{a}'(\bar{\mathbf{x}} - \bar{\mathbf{y}}) - \mathbf{a}'(\mu_1 - \mu_2)] \sim N(0, \mathbf{a}'\Sigma\mathbf{a})$ Why?
- $\mathbf{a}'(n_1 + n_2 - 2)\mathbf{S}_p\mathbf{a} \sim W_p(n_1 + n_2 - 2, \mathbf{a}'\Sigma\mathbf{a})$

- When k is small, Bonferroni's inequation can be used to give $\mathbf{a}_i'(\mu_1 - \mu_2), i=1, 2, \dots, k\}$ $1-\alpha$ simultaneous confidence interval:

$$\mathbf{a}_i'(\bar{\mathbf{x}} - \bar{\mathbf{y}}) - t_{\alpha/2k}(n_1 + n_2 - 2) \sqrt{\frac{n_1 + n_2}{n_1 n_2}} \sqrt{\mathbf{a}_i'\mathbf{S}_p\mathbf{a}_i} \leq \mathbf{a}_i'(\mu_1 - \mu_2) \quad (4.3.7)$$

$$\leq \mathbf{a}_i'(\bar{\mathbf{x}} - \bar{\mathbf{y}}) + t_{\alpha/2k}(n_1 + n_2 - 2) \sqrt{\frac{n_1 + n_2}{n_1 n_2}} \sqrt{\mathbf{a}_i'\mathbf{S}_p\mathbf{a}_i}$$

Why?

To explain (4.3.7), we first notice that

$$\frac{a_i'(\bar{x}-\bar{y})-a_i'(\mu_1-\mu_2)}{\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})a_i'\Sigma a_i}} \sim N(0,1)$$

$$\text{and } \frac{a_i'S_p a_i}{a_i'\Sigma a_i} = \frac{a_i'[(n_1-1)S_1+(n_2-1)S_2]a_i}{(n_1+n_2-2)a_i'\Sigma a_i} = \frac{1}{n_1+n_2-2} \frac{(n_1-1)a_i'S_1 a_i+(n_2-1)a_i'S_2 a_i}{a_i'\Sigma a_i} \\ \sim \chi^2(n_1 + n_2 - 2)/(n_1 + n_2 - 2).$$

Therefore, according to the definition of χ^2 distribution, we have

$$\frac{a_i'(\bar{x}-\bar{y})-a_i'(\mu_1-\mu_2)}{\sqrt{(\frac{1}{n_1}+\frac{1}{n_2})a_i'\Sigma a_i}} / \sqrt{\frac{a_i'S_p a_i}{a_i'\Sigma a_i}} \sim t(n_1 + n_2 - 2).$$

It follows that

$$\frac{a_i'(\bar{x}-\bar{y})-a_i'(\mu_1-\mu_2)}{\sqrt{\frac{n_1+n_2}{n_1 n_2} a_i'S_p a_i}} \sim t(n_1 + n_2 - 2). \quad (4.3.7a)$$

The $1-\alpha$ (Bonferroni's) simultaneous confidence interval based on the above t statistic are given as (4.3.7).

- 
- **Example 4.3.1(Example 4.2.1 continued)** Table 4.3.1 lists the data for 9 two-year-old girls corresponding to Table 4.2.1. We plan to test the existence of significance difference between boys' and girls' mean vectors under the multivariate normality hypothesis.

Table 4.3.1

Physical Measurement Data of Rural Girls

Number	Height(cm)	C. C.(cm)	U.A.G(cm)
1	80	58.4	14.0
2	75	59.2	15.0
3	78	60.3	15.0
4	75	57.4	13.0
5	79	59.5	14.0
6	78	58.1	14.5
7	75	58.0	12.5
8	64	55.5	11.0
9	80	59.2	12.5

➤ From Example 4.2.1:

$$n_1 = 6, \quad \bar{\mathbf{x}} = (82.0, 60.2, 14.5)'$$

$$(n_1 - 1)\mathbf{S}_1 = \begin{pmatrix} 158.00 & 40.20 & 2.50 \\ 40.20 & 15.86 & 6.55 \\ 2.50 & 6.55 & 9.50 \end{pmatrix}$$

From Table 4.3.1:

$$n_2 = 9, \quad \bar{\mathbf{y}} = (76.0, 58.4, 13.5)'$$

$$(n_2 - 1)\mathbf{S}_2 = \begin{pmatrix} 196.00 & 45.10 & 34.50 \\ 45.10 & 15.76 & 11.65 \\ 34.50 & 11.65 & 14.50 \end{pmatrix}$$

So

$$\bar{\mathbf{x}} - \bar{\mathbf{y}} = (6.0, 1.8, 1.0)'$$

$$\mathbf{S}_p = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2} = \begin{pmatrix} 27.2308 & 6.5615 & 2.8462 \\ 6.5615 & 2.4323 & 1.4000 \\ 2.8462 & 1.4000 & 1.8462 \end{pmatrix}$$

$$(4.3.2) \quad T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}} - \bar{\mathbf{y}})' \mathbf{S}_p^{-1} (\bar{\mathbf{x}} - \bar{\mathbf{y}}) = 5.312$$

$$T_{0.05}^2 = \frac{p(n_1 + n_2 - 2)}{n_1 + n_2 - p - 1} F_{0.05}(p, n_1 + n_2 - p - 1)$$

$$T_{\alpha}^2 = \frac{p(n_1 + n_2 - 2)}{n_1 + n_2 - p - 1} F_{\alpha}(p, n_1 + n_2 - p - 1) = \frac{3 \times 13}{11} \times F_{0.05}(3, 11) = \frac{3 \times 13}{11} \times 3.59 = 12.728$$

Since $T^2 < T_{0.05}^2$, H_0 can not be rejected, namely the two mean vectors are not significantly different ($p=0.27$).

Example 4.3.2. There are two brands of tires, brand A and brand B. Currently, six tires are selected for durability testing, which is conducted in three stages with each stage rotating 1000 times. The measured values of durability indicators are listed in the following table

Stages for brand A			Stages for brand B		
1	2	3	1	2	3
194	192	141	239	127	90
208	188	165	189	105	85
233	217	171	224	123	79
241	222	201	243	123	110
265	252	207	244	117	100
269	283	191	226	125	75

Let's ask whether there is a significant difference in durability indicators between brand A and brand B tires under the assumption of multivariate normality with the common covariance matrix between the two populations. If yes, which stage is different. (The confidence level of the test is 0.05)

Solution: Let durability index vector of brand A be $\mathbf{x} = (x_1, x_2, x_3)'$, and durability index vector of brand B be $\mathbf{y} = (y_1, y_2, y_3)'$. Then $\mathbf{x} \sim N_3(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$, $\mathbf{y} \sim N_3(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. Let n_1 the sample size of $\mathbf{x}$ and n_2 be the sample size of $\mathbf{y}$. Based on known conditions, we $n_1 = n_2 = 6$,

$$\begin{aligned}\bar{\mathbf{x}} &= (235, 225.67, 179.33)^T, \bar{\mathbf{y}} = (227.33, 120, 89.83)^T \\ \mathbf{S}_1 &= \begin{pmatrix} 901.2 & 1026.4 & 666.4 \\ 1026.4 & 1324.267 & 644.133 \\ 666.4 & 644.133 & 623.067 \end{pmatrix} \\ \mathbf{S}_2 &= \begin{pmatrix} 421.867 & 128 & 143.667 \\ 128 & 65.2 & -0.6 \\ 143.667 & -0.6 & 174.167 \end{pmatrix} \\ \mathbf{S}_p &= \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1+n_2-2} = \begin{pmatrix} 661.5335 & 577.2 & 405.0335 \\ 577.2 & 694.7335 & 321.7665 \\ 405.0335 & 321.7665 & 398.6170 \end{pmatrix}\end{aligned}$$

Hotelling T^2 is calculated as

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}} - \bar{\mathbf{y}})^T \mathbf{S}_p^{-1} (\bar{\mathbf{x}} - \bar{\mathbf{y}}) = 365.3140$$

Under H_0 , we have $T_\alpha^2 \sim \frac{(n_1+n_2-2)p}{(n_1+n_2-p-1)} F_\alpha(p, n_1 + n_2 - p - 1)$ with $T_{0.05}^2 = 15.2652 < T^2$.

Equivalently, $F = \frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2 = 97.42 > 4.07 = F_{0.05}(3,8)$. Therefore, we can conclude that there is a significant difference in durability indicators between brand A and brand B tires.

Next, we hope to test which stage leads to the difference between μ_1 and μ_2 . For the i -th stage ($i = 1,2,3$), we consider the following hypothesis test:

$$H_{0i}: \mu_{1i} = \mu_{2i} \quad \text{vs} \quad H_{1i}: \mu_{1i} \neq \mu_{2i}.$$

The t test statistic is constructed as

$$t_i = \frac{\bar{x}_i - \bar{y}_i}{\sqrt{(\frac{1}{n_1} + \frac{1}{n_2}) S_{p,(i,i)}}}, \quad \frac{a'_i(\bar{x} - \bar{y}) - a'_i(\mu_1 - \mu_2)}{\sqrt{\frac{n_1+n_2}{n_1 n_2} a'_i S_p a_i}} \sim t(n_1 + n_2 - 2) \quad (4.3.7a)$$

where $S_{p,(i,i)}$ is the (i, i) element of S_p . Under H_{0i} , $t_i \sim t(n_1 + n_2 - 2)$. By calculation, we obtain

$$|t_1| = 0.5162, |t_2| = 6.9437, |t_3| = 7.7644.$$

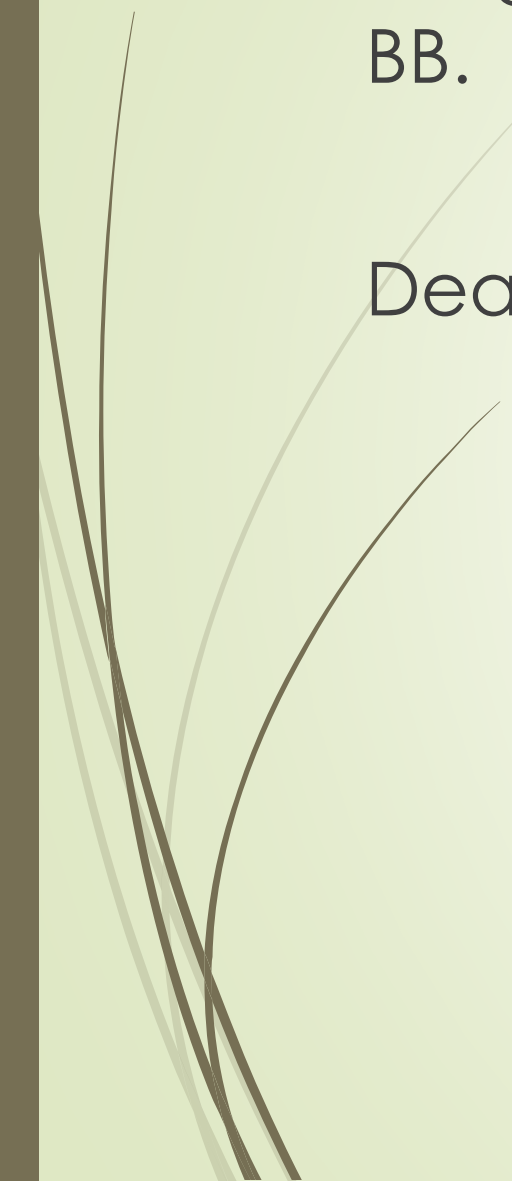
By noticing $t_{0.025, n_1+n_2-2} = 2.2281$, we have $|t_1| < 2.2281, |t_2| > 2.2281, |t_3| > 2.2281$.

Therefore, the second and third stages leads to the difference between brand A and brand B.



Assignments: Homework has been published on BB.

Deadline: 16:20, March 14 (next Thursday), 2025



§ 4.3.2 pairing test T^2 statistic

➡ Suppose $(\mathbf{x}_i, \mathbf{y}_i)$, $i=1, 2, \dots, n (n > p)$ are pairing test data. Let

$$\mathbf{d}_i = \mathbf{x}_i - \mathbf{y}_i, \quad i=1, 2, \dots, n$$

Suppose $\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_n$ are independently and identically distributed to $N_p(\boldsymbol{\delta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} > 0$, $\boldsymbol{\delta} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$, $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ are respectively mean vectors of population $\mathbf{x}$ and $\mathbf{y}$. Hope test

$$H_0: \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2, \quad H_1: \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$$

equivalent to

$$H_0: \boldsymbol{\delta} = \mathbf{0}, \quad H_1: \boldsymbol{\delta} \neq \mathbf{0}$$

In this way, two population mean comparative test can be translated into one population situation. The Hotelling T^2 Statistic :

$$T^2 = n \bar{\mathbf{d}}' \mathbf{S}_d^{-1} \bar{\mathbf{d}}$$



where

$$\bar{\mathbf{d}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}$$

$$\mathbf{S}_d = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{d}_i - \bar{\mathbf{d}})(\mathbf{d}_i - \bar{\mathbf{d}})'$$

when $H_0: \boldsymbol{\delta} = \mathbf{0}$ is true, statistic:

$$\frac{n-p}{p(n-1)} T^2 \sim F(p, n-p)$$

For a given significance level α , rejection rule:

Reject H_0 if $T^2 \geq T_\alpha^2$, where

$$T_\alpha^2 = \frac{p(n-1)}{n-p} F_\alpha(p, n-p)$$


§ 4.4 Single Population Mean Component Structural Relationship Test

- Suppose $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is a sample of multivariate normal population $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} > 0$, $n > p$. Hypothesis Test:

$$H_0: \mathbf{C}\boldsymbol{\mu} = \boldsymbol{\varphi}, \quad H_1: \mathbf{C}\boldsymbol{\mu} \neq \boldsymbol{\varphi} \quad (4.4.2)$$

where $\mathbf{C}$ is a known $k \times p$ matrix, $k < p$, $\text{rank}(\mathbf{C}) = k$ and $\boldsymbol{\varphi}$ is a given k -vector.

On the basis of multivariate normal distribution properties,

$$\mathbf{C}\mathbf{x} \sim N_k(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$$

Since

$$\text{rank}(\mathbf{C}\boldsymbol{\Sigma}\mathbf{C}') = \text{rank} \left[\mathbf{C}\boldsymbol{\Sigma}^{\frac{1}{2}} \left(\mathbf{C}\boldsymbol{\Sigma}^{\frac{1}{2}} \right)' \right] = \text{rank} \left(\mathbf{C}\boldsymbol{\Sigma}^{\frac{1}{2}} \right) = \text{rank}(\mathbf{C}) = k$$

Then $\mathbf{C}\Sigma\mathbf{C}' > 0$. We can construct Hotelling statistic to test (4.4.2), which is given by

$$T^2 = n(\mathbf{C}\bar{\mathbf{x}} - \boldsymbol{\varphi})' (\mathbf{CSC}')^{-1} (\mathbf{C}\bar{\mathbf{x}} - \boldsymbol{\varphi}),$$

where $T^2 \sim T^2(k, n-1)$. When $H_0: \mathbf{C}\boldsymbol{\mu} = \boldsymbol{\varphi}$ is true,

$$\frac{n-k}{k(n-1)} T^2 \sim F(k, n-k)$$

► **Properties:**

$$\frac{n-p+1}{pn} T^2(p, n) \sim F(p, n-p+1)$$

where n is the free degree of the above Wishart Distribution

For a given significance level α , the rejection rule is:

If $T^2 \geq T_{\alpha}^2$, reject H_0 .

where $T_{\alpha}^2 = \frac{k(n-1)}{n-k} F_{\alpha}(k, n-k)$.

► Especially, when the following hypothesis will be tested,

$$H_0: \mathbf{C}\boldsymbol{\mu} = \mathbf{0}, \quad H_1: \mathbf{C}\boldsymbol{\mu} \neq \mathbf{0}$$

then T^2 can be simplified as

$$T^2 = n\bar{\mathbf{x}}' \mathbf{C}' (\mathbf{CSC}')^{-1} \mathbf{C}\bar{\mathbf{x}}$$

This leads to the following profile analysis(轮廓分析)

4.4 Contour analysis (轮廓分析)

► **Example 4.3.1** Suppose $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_p)'$, $\boldsymbol{\Sigma} > 0$, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ is a sample chosen from the above population. Hypothesis test is

$$H_0: \mu_1 = \mu_2 = \dots = \mu_p, \quad H_1: \mu_i \neq \mu_j, \text{ at least one couple } i \neq j,$$

where H_0 implies the profile is horizontal. Let

$$\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & \dots & 0 \\ 1 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -1 \end{pmatrix}$$

then the above hypothesis is expressed as:

$$H_0: \mathbf{C}\boldsymbol{\mu} = \mathbf{0}, \quad H_1: \mathbf{C}\boldsymbol{\mu} \neq \mathbf{0}.$$

The test statistic is

$$T^2 = n\bar{\mathbf{x}}' \mathbf{C}' (\mathbf{C} \mathbf{S} \mathbf{C}')^{-1} \mathbf{C} \bar{\mathbf{x}},$$

where $T^2 \sim T^2(p-1, n-1)$.

§ 4.4 轮廓分析

设对同一个单元(个人、商店、小块土地等)施加 p 种处理(如测验,问卷调查等)或在相继 p 个时间段内重复测量,依次得到测量值 $x_1, x_2, \dots, x_p$, 其相应的均值依次为 $\mu_1, \mu_2, \dots, \mu_p$ 。

图 4.4.1 是将点 $(1, \mu_1), (2, \mu_2), \dots, (p, \mu_p)$ 用直线连接起来的折线图,称为总体的轮廓(profile)。轮廓分析是轮廓的分析或多个轮廓的比较,它常用于分析比较相继一连串的心理测试或其他测试中。我们这里讨论单总体(或单组)和两总体(或两组)的轮廓分析。

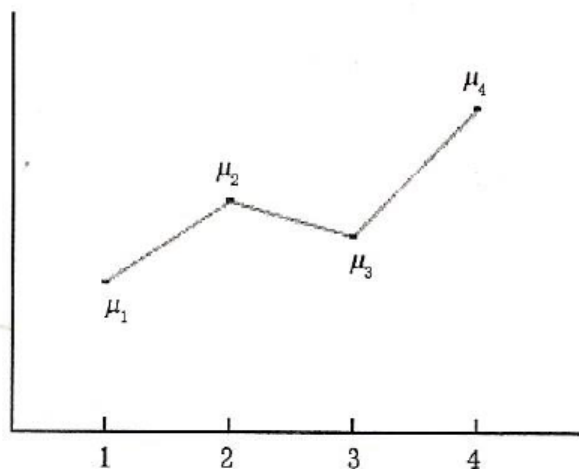


图 4.4.1 总体轮廓($p=4$)

一、单总体的轮廓分析

为了比较均值 $\mu_1, \mu_2, \dots, \mu_p$, 一个基本的假设为轮廓是水平的, 即

$$H_0: \mu_1 = \mu_2 = \dots = \mu_p, \quad H_1: \mu_i \neq \mu_j, \text{ 至少存在一对 } i \neq j \quad (4.4.1)$$

For a given significance level α , the rejection rule is :

If $T^2 \geq T_\alpha^2$, reject H_0

where $T^2 \sim T^2(p-1, n-1)$ and

$$T_\alpha^2 = \frac{(p-1)(n-1)}{n-p+1} F_\alpha(p-1, n-p+1)$$

► **Properties :**

$$\frac{n-p+1}{pn} T^2(p, n) \sim F(p, n-p+1)$$

where n is the free degree of the above Wishart Distribution

- Since $\mathbf{C}$ is row full rank and every row is a comparison vector (which have one “1” and one “-1” elements and others are 0) , $\mathbf{C}$ is called comparison matrix(对比矩阵) .
- In this example, comparison matrix $\mathbf{C}$ is not unique, like

$$\mathbf{C}^* = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{pmatrix}$$

- **Example 4.3.2.** In Example 4.2.1, assume human has such a universal law that the average dimensional proportion of high and chest circumference and upper arm girth is 6:4:1. We hope the data of Table 4.2.1 conform to the above regulation. Then we need to test

$H_0: \mu_1/6 = \mu_2/4 = \mu_3, H_1: \mu_1/6, \mu_2/4, \mu_3$ at least one couple are not equal

Let

$$C = \begin{pmatrix} 2 & -3 & 0 \\ 1 & 0 & -6 \end{pmatrix}$$

Then the above hypothesis can be expressed as

$$H_0: C\mu = 0, H_1: C\mu \neq 0$$

$$S = \begin{pmatrix} 31.600 & 8.040 & 0.500 \\ 8.040 & 3.172 & 1.310 \\ 0.500 & 1.310 & 1.900 \end{pmatrix}$$

After calculation,

$$C\bar{x} = \begin{pmatrix} -16.6 \\ -4.0 \end{pmatrix}, CSC' = \begin{pmatrix} 58.468 & 56.660 \\ 56.660 & 94.000 \end{pmatrix}$$

Therefore,

$$(CSC')^{-1} = (2285.6364)^{-1} \begin{pmatrix} 94.000 & -56.660 \\ -56.660 & 58.468 \end{pmatrix}$$

So

$$T^2 = n\bar{\mathbf{x}}'\mathbf{C}'(\mathbf{CSC}')^{-1}\mathbf{C}\bar{\mathbf{x}} = 6 \times 8.450 = 50.700$$

Since the above $T^2 \sim T^2(k, n - k)$, we have

$$T_{0.01}^2 = \frac{k(n-1)}{n-k} F_{0.01}(k, n-k) = \frac{2 \times 5}{4} \times 18.0 = 45 < T^2$$

we reject H_0 , namely this group of data is not consistent to human universal law ($p=0.008$).

➤ The above $\mathbf{C}$ can be chosen as

$$\mathbf{C}^* = \begin{pmatrix} 0 & 1 & -4 \\ 1 & 0 & -6 \end{pmatrix}$$

and the result remains unchanged.



Assignment: Pls find the assignment file in BB.

